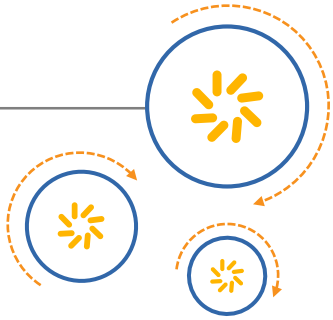
Qualcomm Technologies, Inc.



High Voltage Dedicated Charging Port

Qualcomm® Quick Charge™ 3.0 Interface Specification

80-NH008-2 Rev. J

February 23, 2016

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Revision history

Revision	Date	Description
AA	July 2013	Initial release
AB	September 2013	Added tolerance for the 200 mV specification, allowed continuous for Class B, and updated the flow chart
B	May 2014	Section 1.1 HVDCP specification: Updated the minimum voltage for Class A from 5 V to 3.6 V and for Class B from 5 V to 3.6 V Figure 1-2 HVDCP wall adapter functional flowchart: Updated the flowchart Figure 1-3 Portable device functional flowchart: Updated the flowchart Table 2-1 DC operating characteristics: Updated minimum and maximum values of VBUS and HV from -5 and 5 to -10 and 10 respectively Updated VBUS, CONT, and RANGE minimum values from 5 to 3.6 Updated parameters for TGLITCH_V_CHANGE
C	June 16, 2014	Figure 1-2 HVDCP wall adapter functional flowchart: Updated D- < Vsel_ref, 2 V to D- > Vsel_ref, 2 V
D	June 24, 2014	Figure 1-2 HVDCP wall adapter functional flowchart: Correction made to the condition D- < Vsel_ref, 2 V to D- > Vsel_ref, 2 V in order to meet D+ = D- = 0.6 V for decreasing V_{OUT}
E	August 2014	Figure 1-2 HVDCP wall adapter functional flowchart: Updated the figure for continuous mode implementation Figure 1-3 Portable device functional flowchart: Updated the figure for continuous mode implementation Table 2-1 DC operating characteristics: Added operating characteristics for $T_{GLITCH_V_CHANGE}$, $T_{GLITCH_CONT_CHANGE}$, T_{ACTIVE}, $T_{INACTIVE}$, $T_{V_NEW_REQUEST_CONT}$, $T_{V_CONT_CHANGE_SINGLE}$, and $T_{V_CONT_CHANGE_GROUP}$ Section 3.3 Continuous mode of operation (Quick Charge 2.1): Added section Figure 3-1 Continuous mode of operation example timing diagram: Added figure Figure 3-2 HVDCP recommended operating range: Added figure
F	January 2015	Figure 1-3 Portable device function flowchart: Changed D- from Vdm_src to Vdm_up for the decrement by 200 mV case Figure 3-1 Continuous mode of operation example timing diagram: Changed Tv_cont_change_single to start when D+/- change state Changed Tv_cont_change_group to start when D+/- change state
G	February 2015	Global: Updated Quick Charge 2.1 references to Quick Charge 3.0 throughout the document Section 3.3 Continuous mode of operation (Quick Charge 3.0): Added limitation information Figure 8 Continuous mode of operation example timing diagram: Changed and clarified the start time of VBUS changing after an increment or decrement is requested

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Revision	Date	Description
H	October 2015	▪ Re-organized all of the information for improved clarity so did not use change bars ▪ Created separate tables for timing and static parameters ▪ Added additional reserved state to Table 1 HVDCP voltage modes ▪ Added reference documentation ▪ Added definition of terms Table 2 Definition of terms ▪ Added timing diagrams for voltage requests ▪ Removed grouped pulse description ▪ Modified definition of $T_{V_CONT_CHANGE}$ to accommodate both grouped and single pulse implementations ▪ Added power profile design considerations ▪ Added VBUS collapse considerations
Revision I was omitted in accordance with QTI document conventions.		
J	February 2016	▪ Section 1.1 Quick Charge overview: Added 18 W minimum for QC3.0 adapters by requiring at least 6 V/3 A ▪ Table 1 HVDCP voltage modes: Removed the 3.3 V High-Z portable device signal ▪ Updated Section 4.1 Power profile design and Section 4.2 Adapter collapse behavior ▪ Updated Figure 7 Example Class A HVDCP power profile for HVDCPs and Figure 8 Example Class A HVDCP constant power profile ▪ Updated Table 4 VBUS collapse thresholds voltage in continuous mode to 4.25 V ▪ Section 4.2 Adapter collapse behavior: Added the 200 mA minimum load requirement for regulation below 5 V ▪ Section 4.5 Battery powered devices: Added stipulations for battery powered devices supporting continuous mode

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1 Introduction

This specification provides the requirements for adapters to meet the Quick Charge 3.0 standard. Adapters that meet these requirements are referred to as high-voltage dedicated charging ports (HVDCPs). An HVDCP interoperates with Quick Change 3.0-enabled portable devices, as well as maintains backwards compatibility with Quick Charge 2.0- and Quick Charge 1.0-enabled portable devices.

1.1 Quick Charge overview

The USB BC1.2 specification defines a class of adapter knows as a dedicated charging port (DCP). These adapters can provide currents up to 1.8 A, but are limited to a voltage of 5 V. To achieve high charge rates in portable devices, Quick Charge 2.0 was introduced to overcome cable resistance voltage drops and connector current ratings by enabling a high-voltage charging specification while leveraging USB BC1.2 compliant signaling. This signaling allows a portable device to request a voltage greater than 5 V, based on Table 1. There are two classes of HVDCPs, Class A and Class B. A Class A HVDCP must provide voltage outputs of 5 V, 9 V, and 12 V, while a class B HVDCP must provide 20 V in addition to the Class A requirements. A Quick Charge 3.0-compliant HVDCP support a minimum of 18 W by providing 3 A at a 6 V output voltage. The Quick Charge 3.0-compliant HVDCP may increase its power rating by increasing the voltage at which 3 A is supported.

Table 1 HVDCP voltage modes

Portable device signal		HVDCP output
D+	D-	Adapter voltage
0.6 V	0.6 V	12 V
3.3 V	0.6 V	9 V
0.6 V	3.3 V	Continuous mode
3.3 V	3.3 V	20 V
0.6 V	High-Z	5 V (DCP)

An HVDCP leverages the previously reserved state of the HVDCP voltage modes, now referred to as continuous mode, to enable a Quick Charge 3.0-enabled portable device to dynamically increment or decrement the voltage on VBUS in 200 mV discrete steps. These increment and decrement requests are made by issuing a pulse on D+ or D- from the static continuous mode USB signal levels. The voltage range of requests in continuous mode extends from 3.6 V to 12 V for Class A and 20 V for Class B HVDCPs. This 200 mV resolution on VBUS voltage request allows the portable device to manage power dissipation and thermal performance while charging.

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1.2 References

Table 2 References

Name and document number	Type
Universal Serial Bus, Battery Charging Specification, Revision 1.2	USB BC 1.2
Universal Serial Bus, Type-C Cable and Connector Specification, Revision 1.1	USB type-C
High Voltage Dedicated Charging Port Quick Charge 2.0 Interface Specification (80-NH008-1)	Qualcomm Technologies, Inc. (QTI)
Quick Charge over USB Type-C Application Note (80-NH008-13)	QTI

1.3 Definition of terms

Table 3 defines the terms used in this document that are not explicitly defined in the reference documents listed in Table 2. The descriptions are for the purposes of this document and not intended to be used in other contexts.

Table 3 Definition of terms

Term	Description
HVDCP	High-voltage dedicated charging port. A power source complaint with the Quick Charge specification for voltage negotiation.
Portable device	Any device capable of requesting Quick Charge compliant voltage requests.
Glitch filter	The time required for a signal to remain across a given threshold after which the signal is considered valid.

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2 Hardware signaling requirements

An HVDCP leverages USB BC 1.2-compliant signaling on D+ and D- to negotiate voltage requests on VBUS with a Quick Charge-enabled portable device. Figure 1 shows the basic blocks required to support the Quick Charge specification.

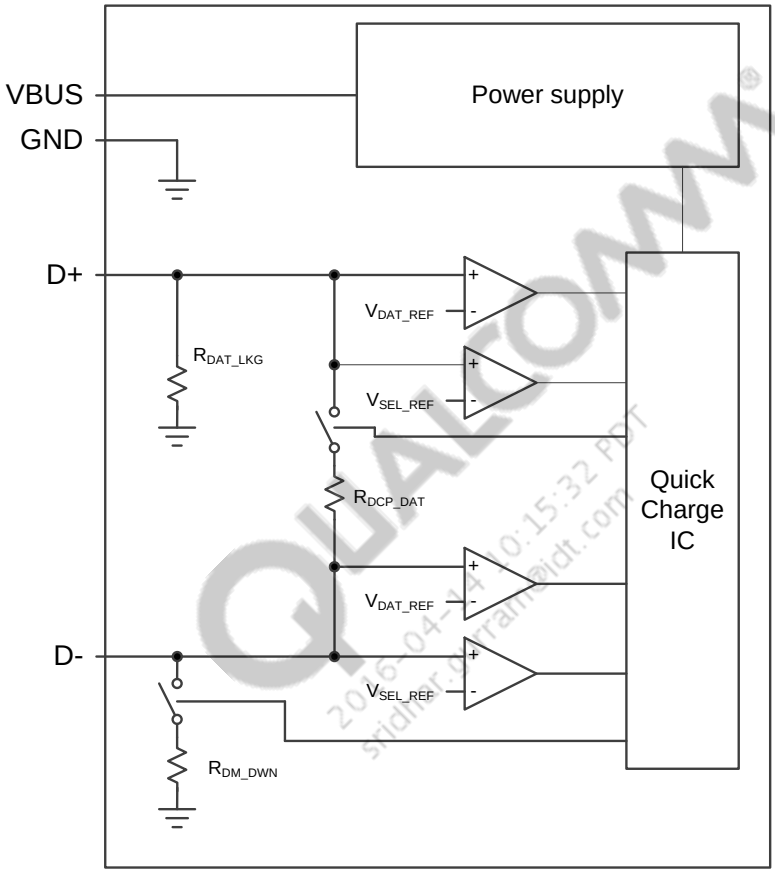


Figure 1 HVDCP block diagram

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3 HVDCP operation

3.1 HVDCP detection

An HVDCP behaves as a standard DCP for a non-Quick Charge-enabled portable device per the USB BC 1.2 specification. Figure 2 shows a high-level flow chart of this detection. At power-up, the adapter shorts D+ and D- through R_{DCP_DAT} within $T_{D+_D-_SHORT}$ of the HVDCP output rising above $V_{OTG_SESS_VLD}$ to allow the portable device to complete USB BC 1.2 detection of a DCP. When D+ is biased above V_{DAT_REF} during USB BC1.2 detection, the $T_{GLITCH_BC_DONE}$ glitch filter ensures that the portable device can complete USB BC 1.2. At the expiration of this glitch filter, the HVDCP opens circuit D+ and D- and applies R_{DM_DWM} to signal to the portable device that the adapter supports Quick Charge. The HVDCP ensures that D- remains below V_{DAT_REF} for $T_{GLITCH_DM_LOW}$ before responding to a voltage mode request based on Table 1. After the expiration of $T_{GLITCH_DM_LOW}$, the HVDCP changes voltage modes within $T_{GLITCH_MODE_CHANGE}$ upon receiving a new voltage request and apply R_{DM_DWM}. If the portable device requests a voltage mode that the HVDCP does not support, then the HVDCP will continue to output the previously requested voltage mode on VBUS. The HVDCP will maintain R_{DM_DWM} until an unplug is detected. See Section 4.1 for details about portable device unplug detection.

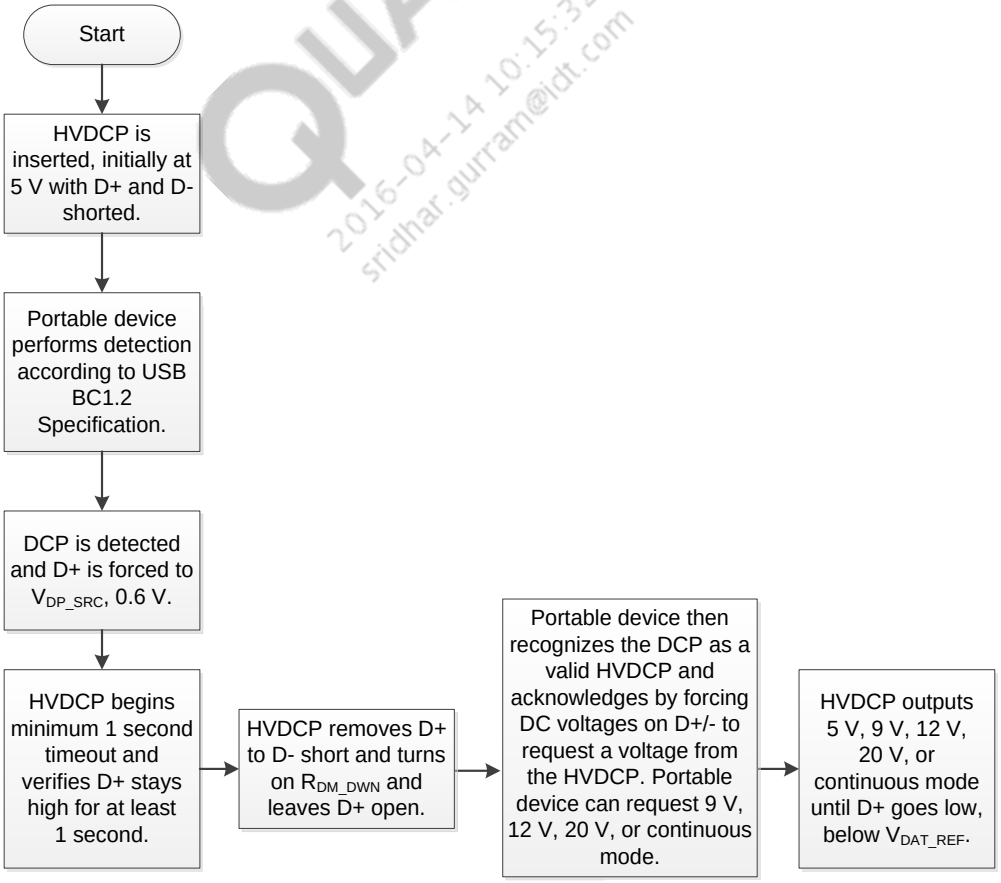


Figure 2 HVDCP detection flowchart

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3.2 Quick Charge 2.0 requests

After the HVDCP has detected D- below V_{DAT_REF} for $T_{GLITCH_DM_LOW}$, the HVDCP accepts the voltage mode requests, based on Table 1, from the portable device. The HVDCP uses V_{SEL_REF} to differentiate between the 0.6 V and 3.3 V values in Figure 3 that correspond to USB BC 1.2-compliant voltage biasing V_{SRC} and V_{UP}. Upon receiving a new voltage mode request on D+ and D-, the HVDCP glitch filters this request by $T_{GLITCH_MODE_CHANGE}$ before entering a new voltage mode. Once the HVDCP has entered a new voltage mode, the HVDCP responds to the Quick Charge 2.0 requests within $T_{V_NEW_REQUEST}$ by changing its VBUS output to within V_{BUS_5V} or V_{BUS_HV} for the requested voltage mode. The $T_{V_NEW_REQUEST}$ time starts simultaneously with $T_{GLITCH_MODE_CHANGE}$ when the request is first made.

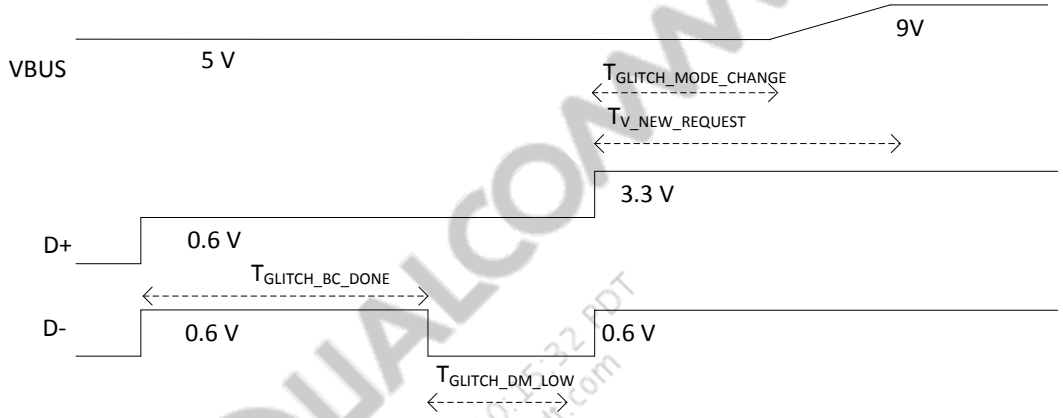


Figure 3 Example request for 9 V mode

3.3 Quick Charge 3.0 requests

A Quick Charge 3.0-enabled portable device may issue a voltage mode request for continuous mode using same approach as a Quick Charge 2.0 voltage mode described in Section 3.2. Once the HVDCP has entered continuous mode after the $T_{GLITCH_MODE_CHANGE}$ glitch filter, as shown in Figure 4, the HVDCP responds to the voltage increment and decrement requests according to a new set of timing requirements, rather than $T_{V_NEW_REQUEST}$.

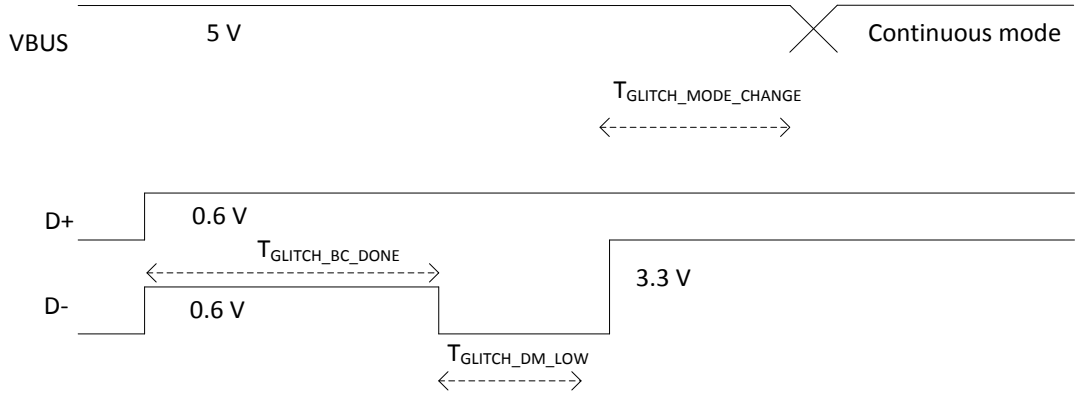


Figure 4 Request for continuous mode

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While in the continuous mode state, the adapter monitors D+ and D- for increment or decrement commands. Increment and decrement commands are signaled as a series of one or more pulses on either D+ or D- to adjust VBUS up or down by $V_{BUS_CONT_STEP}$ for each pulse. The HVDCP processes an increment request as a pulse on D+ above VSEL_REF glitch filtered by $T_{GLITCH_CONT_CHANGE}$ for a time T_{ACTIVE}. A decrement request is processed as a pulse on D- to below V_{SEL_REF} glitch filtered by $T_{GLITCH_CONT_CHANGE}$ for a time $T_{INACTIVE}$.

The HVDCP responds to an increment or decrement by adjusting VBUS by $V_{BUS_CONT_STEP}$ up or down for each pulse received. Independent of the number of pulses received in a series, the voltage on VBUS must settle within the tolerance of $V_{BUS_CONT_STEP}$ for each step within $T_{V_CONT_CHANGE}$ from the rising edge of the last pulse in the series, as shown in Figure 5 and Figure 6. While in continuous mode, the adapter monitors D+ and D- to transition within $T_{GLITCH_MODE_CHANGE}$ to the 5 V voltage mode in Table 1 and change its output voltage within $T_{V_NEW_REQUEST}$.

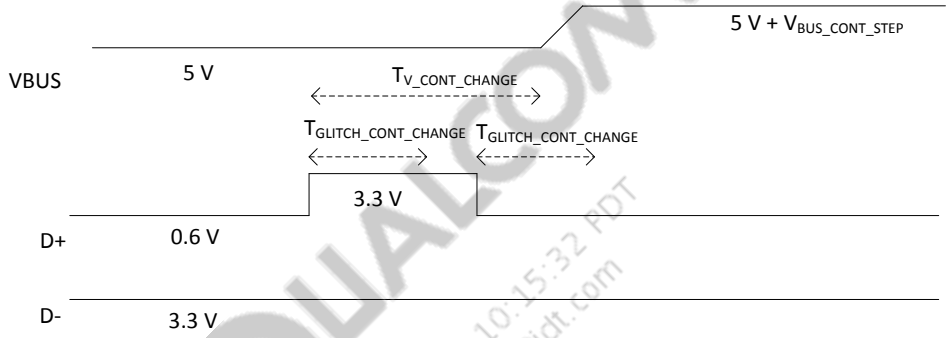


Figure 5 Example of single pulse in continuous mode

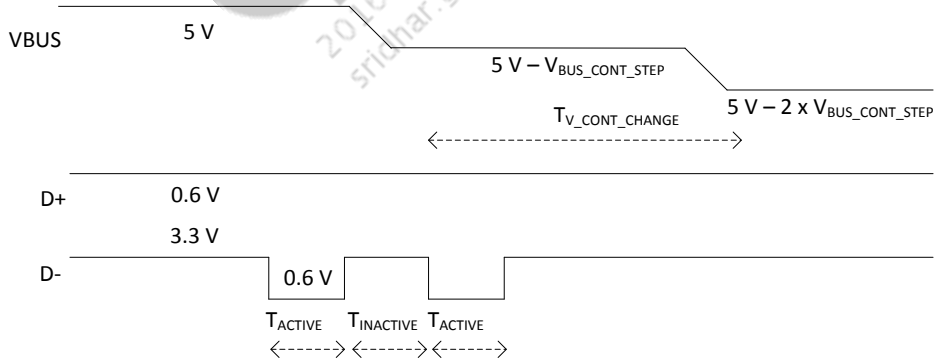


Figure 6 Example of multiple pulses in continuous mode

An HVDCP or portable device supporting Quick Charge 3.0 must also ensure that VBUS is reserved only for providing power or performing negotiations via the USB power delivery revision 3.0 specification. All other forms of VBUS negotiation are explicitly prohibited. Similarly, if a type C connector is used, the CC pin can only negotiate a higher voltage than 5.5 V if USB power delivery is used. Due to the high speeds required on D+ and D- for Quick Charge 3.0 and to ensure there are no interoperability challenges, D+ and D- may only be used for functions listed in the USB BC 1.2 specification or USB 2.0 specification. All other uses besides Quick Charge 2.0 or Quick Charge 3.0 of D+ and D- for negotiating a voltage on VBUS are prohibited.

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4 VBUS considerations

4.1 Power profile design

The power profile of an HVDCP requires careful consideration of cost, interoperability, and safety. Adherence to these guidelines is required to ensure safe and robust operation with Quick Charge-enabled portable devices. Figure 7 and Figure 8 show a visual representation of the concepts described in this section.

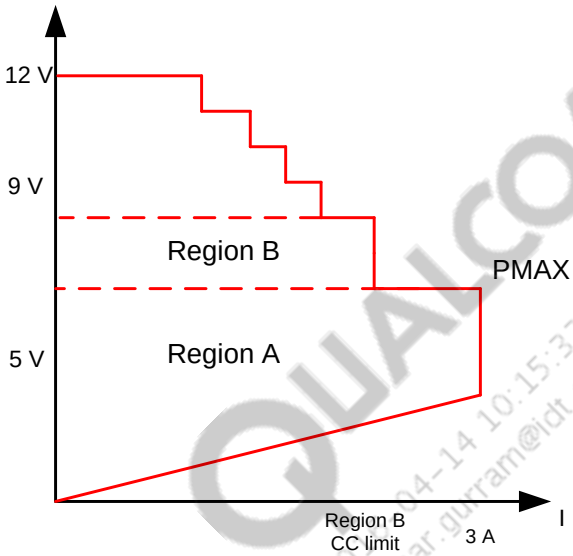


Figure 7 Example Class A HVDCP power profile for HVDCPs

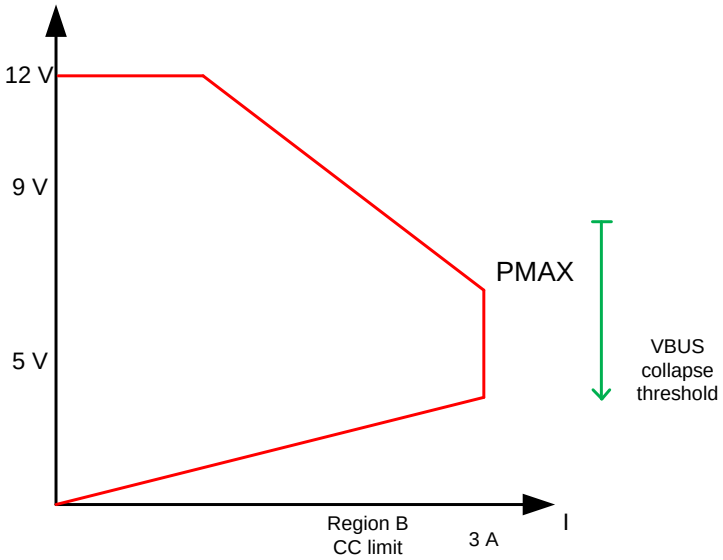


Figure 8 Example Class A HVDCP constant power profile

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For Quick Charge 2.0 modes of operation, discrete constant current levels are sufficient to support the individual voltage modes. This allows flexibility for power providers to select a power profile that is warranted for a wide variety of markets. This same flexibility extends for a Quick Charge 3.0-enabled HVDCP, with the additional requirement that the HVDCP must support 3 A from 3.6 V to 6 V. In order to allow for flexible HVDCP power design, the HVDCP may reduce the current output rating as voltage increases above 6 V to limit the power rating and transition from Region A to Region B. Important guidelines must be followed around the maximum power point of Region A noted at the PMAX point.

As the HVDCP receives increment requests starting at the 5 V and 3 A in Region A, the HVDCP output operating point reaches PMAX. At this point PMAX, the HVDCP may reduce its constant current rating such that the adapter output characteristic operates in Region B. Alternatively, the HVDCP may maintain a constant power output as shown in Figure 8. At the transition from PMAX to Region B, the HVDCP may behave in one of two ways to indicate that it has reached the power limit of Region A or PMAX. The HVDCP may not respond to voltage increment requests while it is limited by its constant current regulation loop in Region A, or the HVDCP may reduce its constant current output to Region B CC limit before increasing its output voltage and operating in Region B. Once the HVDCP is operating in Region B, the HVDCP must allow the portable device to decrease the voltage requested to operate in Region A and increase its constant current limit back to 3 A.

To gain the most benefit from Quick Charge 3.0, the HVDCP power profile will be designed with the maximum possible current so that point A is at a relatively low voltage for the same power level. For example, 6 V/3 A is a more desirable operating point than 9 V/2 A for a portable device.

4.2 Adapter collapse behavior

In order to guarantee interoperability with Quick Charge enabled portable devices, the HVDCP must allow VBUS to collapse with specific behavior requirements in order to allow accurate detection of the adapter current limit. The HVDCP protection schemes must be designed with these criteria in mind in order to prevent a poor charging experience for the consumer.

When the adapter constant current limit rating is exceeded, the adapter output voltage must be allowed to collapse below a certain threshold in order for the portable device to detect the adapter current limit successfully as seen in Figure 9. The HVDCP must allow the voltage on VBUS to remain below the given threshold for T_{GLITCH_UVLO} in order to allow the portable device to detect that its current limit has been reached. While the voltage is collapsed below the noted threshold, the HVDCP must not change voltage modes or reduce output current.

While regulating voltage below 5 V, the adapter must be able to regulate down to 3.6 V at the full 5 V rated current without tripping a protection mechanism. The adapter must also be able to regulate down to 3.6 V with a minimum load of 200 mA.

When the HVDCP current rating is exceeded in Region B, the adapter output voltage will behave in one of these two ways:

- Allow VBUS to collapse down to the same threshold as specified for Region A in the paragraph above.
- Allow VBUS to collapse to its PMAX point at 3 A output current.

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For Quick Charge 2.0 voltage mode requests, VBUS must be allowed to collapse down to at least the thresholds listed in Table 4. While in continuous mode for Quick Charge 3.0 operation below 5 V, the voltage must be allowed to collapse to below 4.25 V, as shown in Figure 9.

Table 4 VBUS collapse thresholds

Nominal voltage	VBUS collapse threshold
5 V	4.25 V
9 V	7.2 V
12 V	9.6 V
Points A and B in continuous mode (if option 1 is selected)	4.25 V

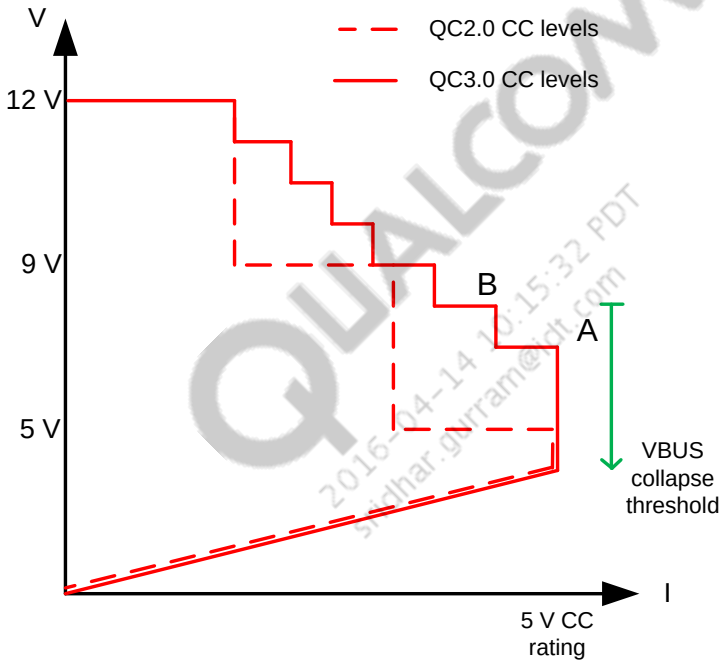


Figure 9 Continuous mode voltage collapse

4.3 Portable device removal

An HVDCP will determine that the portable device has been removed by monitoring the voltage on D+. When D+ falls below V_{DAT_REF} the adapter must re-apply R_{DCP_DAT} and remove R_{DM_DWM} within $T_{D+_D-_SHORT}$ and bring VBUS within V_{BUS_5V} within T_{V_UNPLUG}. If the HVDCP has a type-C output, the HVDCP must discharge VBUS to V_{Safe5V} within $T_{V_UNPLUG_C}$ and to V_{Safe0V} within T_{Safe0V}.

If the HVDCP is designed to operate over a USB type-C connector using a Type-Q to Type-C cable, the CC pins may be exposed to a VBUS through Rp during certain corner conditions. The HVDCP must be able to accommodate this condition by making the CC pins high-voltage tolerant or providing a low-cost external component recommendation. Refer to the *Quick Charge Over USB Type-C Application Note* (80-NH008-13) for further details.

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4.4 Maximum slew rate

The maximum slew rate on VBUS in any condition is limited to V_{SLEW_MAX} in any mode of operation. This applies to both rise edge and falling edge voltage transitions between voltage modes and voltage increment/decrement requests.

4.5 Battery powered devices

Battery powered portable devices, such as juice packs that support Quick Charge 3.0 on an output port, do respond to decrement requests to regulate VBUS below 5 V.

5 Operating parameters

Table 5 DC operating characteristics

Symbol	Description	Min	Typ	Max	Unit
V_{BUS_5V}	Wall adapter output voltage tolerance in 5 V voltage mode	4.75	5	5.5	V
V_{BUS_HV}	Wall adapter output voltage tolerance in 9 V, 12 V, and 20 V voltage modes	-10	–	+10	%
$V_{BUS_CONT_RANGE}$	Class A wall adapter range	3.6	–	12	V
	Class B wall adapter range	3.6	–	20	V
$V_{BUS_CONT_STEP}$	Voltage tolerance per step in continuous mode	0.15	0.2	0.25	V
V_{DAT_REF}	Threshold for D+ to detect the presence of the portable device	0.25	0.325	0.4	V
V_{SEL_REF}	Voltage threshold between V_{UP} and V_{SRC} when selecting a voltage mode	1.8	2	2.2	V
$V_{OTG_SESS_VLD}$	Output voltage at which D+ to D- must be shorted during HVDCP power-up	0.8	–	4.0	V
V_{SLEW_MAX}	Maximum slew rate on VBUS	–	–	30	mV/µs
I_{HVDCP_MIN}	All HVDCPs must output this current at a minimum under all voltage ranges.	500	–	–	mA
R_{DAT_LKG}	Data line leakage	300	900	1500	kΩ
R_{DCP_DAT}	D+ to D- resistance during DCP mode	–	20	40	Ω
R_{DM_DWM}	D- pull-down resistance	14.25	19.53	24.8	kΩ
C_{DCP_PWR}	Equivalent capacitance on D+ and D- to GND	–	–	1	nF

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Table 6 AC operating characteristics

Symbol	Notes	Min	Typ	Max	Unit
$T_{GLITCH_DM_LOW}$	Time D- must remain below V_{DAT_REF} after D+ and D- are open circuited before allowing voltage mode requests.	1	–	–	ms
$T_{GLITCH_BC_DONE}$	Time D+ must remain above V_{DAT_REF} before advertising as HVDCP by open circuiting D+ and D-.	1	–	1.5	sec
$T_{GLITCH_MODE_CHANGE}$	Glitch filter on voltage mode requests	20	40	60	ms
$T_{GLITCH_CONT_CHANGE}$	Rising and falling edge glitch filter on pulses in continuous mode	100	–	200	µs
T_{ACTIVE}	Active pulse time in continuous mode	0.2	–	15	ms
$T_{INACTIVE}$	Time between pulses in continuous mode	200	–	–	µs
$T_{V_NEW_REQUEST}$	Time for an HVDCP change its output voltage to within V_{BUS_5V} or V_{BUS_HV} from the time the voltage mode request is made	–	–	200	ms
$T_{V_CONT_CHANGE}$	Time for an HVDCP change its output voltage to within $V_{BUS_CONT_STEP}$ for each pulse from the rising edge of the last pulse received [1]	–	2	60	ms
T_{GLITCH_UVLO}	Time HVDCP must allow VBUS to collapse below the threshold	20	–	–	ms
T_{V_UNPLUG}	Time for VBUS to discharge to V_{BUS_5V} when D+ goes below V_{DAT_REF}	–	–	500	ms
$T_{V_UNPLUG_C}$	Time for VBUS to discharge to V_{BUS_5V} when D+ goes below V_{DAT_REF} for a USB Type-C output	–	–	275	ms
$T_{D+_D-_SHORT}$	Time for D± to short on HVDCP on unplug when D+ goes below V_{DAT_REF}	–	–	20	ms

1. $T_{V_CONT_CHANGE}$ must target the typical time noted in order to add value and future compatibility.

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